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Perceptions or behavior? An evaluation of CFPB's financial well-being scale using household financial ratios

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Abstract

This article creates a typology to assess four categories of financial well-being based on a combination of household financial ratios. Most financial well-being scales are based on subjective measures (i.e., perceptions), as objective markers have not reliably encapsulated financial well-being. We define a conceptual model, the equilibrium model of the household (EMH), and use discriminant analysis to extract categories of financial well-being. The continuum of these categories is financially distressed (lowest), financially fragile, financially stable, and financially flourishing (highest). Our results demonstrate these categories are consistent with the Consumer Financial Protection Bureau's (CFPB) Financial Well-Being Scale, a subjective scale. Higher CFPB scores were associated with higher category ranks. Our results provide additional evidence to support construct validity of the CFPB scale and may offer more actionability to the CFPB scores because specific financial outcomes/behaviors associated with our categories of financial well-being correspond to ranges of the CFPB scale. However, we argue that the claim whereby the CFPB scale measures a concept beyond traditional financial measures is imprecise and may even reflect the existence of noise in the CFPB's data, raising questions about its reliability.

KEYWORDS

financial flourishing, financial well-being scale, objective measures of well-being, subjective measures of well-being

1 | INTRODUCTION

In 2015, the Consumer Financial Protection Bureau (CFPB) introduced their Financial Well-Being Scale, a simple, 10-item survey to gauge the effectiveness of their efforts on improving the financial lives of the American households (Consumer Financial Protection Bureau, 2015). The development of a financial well-being

scale by the CFPB, while not the first attempt of its kind, marked a significant milestone in the efforts to measure the effectiveness of financial interventions aimed at improving the financial well-being of United States households (Consumer Financial Protection Bureau [CFPB], 2015). Ranging from financial education workshops, to accessing capital or incentivized savings programs, to name a few, financial interventions are

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implemented mainly by public or community-based organizations. Many of the community-based organizations or other entities in the United States that are involved in these efforts currently utilize the CFPB's Financial Well-Being Scale.

The CFPB scale does not measure financial well-being based on financial data. Respondents are not asked about their level of income, assets, and liabilities or rate of savings (indicators henceforth collectively referred to in this article as objective measures). In fact, the developers of the CFPB scale claim that it measures something fundamentally different than what objective indicators of financial status could potentially measure. To make their point, researchers refer to the high standard deviation of the distribution of scores in the national sample to claim that none of the objective measures can independently determine financial well-being (CFPB, 2017). In other words, since any objective measure, such as income, expenses, savings, and so forth, does not perfectly correlate to financial well-being, the CFPB scale offers a better, or more complete, picture of financial well-being. This idea implies that individuals or households with identical objective characteristics may have different perceptions regarding their level of financial well-being (Nanda & Banerjee, 2021). Among other things, in this article, we attempted to evaluate this claim.

2 | LITERATURE REVIEW

Differences between perceptions and behavior, the latter primarily measured through objective indicators, are well recognized in the literature related to financial well-being. Financial well-being is often defined as a composite measure that includes both subjective (perceptive) and objective (behavioral) factors (CFPB, 2015; Gutman et al., 2015; Gutter & Copur, 2011; Joo, 2008; Robb & Woodyard, 2011). There are some minor differences in existing definitions. For example, Joo (2008) preferred to use the term financial wellness rather than financial well-being because the former encompasses a more functional rather than cognitive definition and implied a multidisciplinary application. On the other hand, the Center for Financial Services Innovation (CFSI) used the term financial health in describing a financial behavior that increases the financial resilience and opportunity of individuals and households (Gutman et al., 2015).

Yet, while defined as a measure that includes both objective and subjective components, when actually analyzed, almost all existing scales or tools are based solely on subjective measures (Netemeyer et al., 2018; Prawitz et al., 2006; Sorgente & Lanz, 2019). One exception is the CFSI's Financial Health Segments where objective

measures are included to some extent. However, these variables are measured categorically and are not extracted from detailed household-level financial data (Gutman et al., 2015). Other studies appear to adopt a similar approach. Robb and Woodyard (2011), studied the relationship between financial knowledge and financial satisfaction in predicting best practice financial behavior based on Joo's financial wellness framework. Gutter and Copur (2011) introduced a subjective scale, the In Charge Financial Distress/Financial Well-Being (IFDFW) for college students by including objective measures of financial behavior in the form of a few binary variables.

To the authors' knowledge, since the publication of the aforementioned CFPB (2017) report, with a few notable exceptions, neither the CFPB's Financial Well-Being Scale nor any other financial well-being scale, directly compares or contrasts to a complete set of household financial ratios. A recent study, heavily influenced by the CFPB scale, combined self-reported measures of subjective financial well-being with data from bank records to create a separate "observed" scale of financial well-being for Australian households (Comerton-Forde et al., 2022). The study concluded that both the self-reported subjective scale and the "observed" financial well-being scale maintained a high level of consistency in discriminating between a wide range of financial outcomes. Most notably, the study identified saving behavior as the single, most important predictor of financial well-being on both scales. That being said, there were some differences between the two scales. For example, the observed scale incorporated temporal aspects of financial well-being, such as considering the number of months within a 12-month period in which household expenses exceeded 80% of incoming contributions. The subjective measure, on the other hand, did not include temporal dimensions of financial well-being. The study also found some differences between the self-reported vs. observed scales in terms of their association with certain predictors of financial well-being (Comerton-Forde et al., 2022, p. 147).

Using the CFPB's Financial Well-Being Scale's national dataset, another recent study used structural equations modeling to estimate the effect of indicators of financial capability including savings, financial shocks, income volatility, as well as demographic variables on a score on the CFPB scale (Anvari-Clark & Ansong, 2022). The authors concluded that factors within the control of the household, such as saving behavior, had a substantial predictive influence on financial well-being as compared with structural or environmental variables. In particular, results from their study indicated that when saving increased by one standard deviation, a CFPB score increased by 0.437 standard deviations.

These, and similar studies, incorporated elements of household behavior and contributed to our knowledge of financial well-being. However, overall, little effort has been made to develop a financial well-being scale that combines both types of data (e.g., objective and subjective data). This, in part, is due to the fact that a conjoined theoretical model of objective and subjective measures has yet to be developed by researchers in the field of financial well-being. This issue notwithstanding, it is also important to note that collecting data using objective measures is often more complicated than subjective measures. Direct observation of behavior or collection of relevant objective data is almost impossible to obtain or very costly to administer. On the other hand, self-reported financial data often involves inconsistencies and unsystematic errors. Therefore, several years after the introduction of the CFPB's Financial Well-Being Scale, the gap in the literature on the level of correspondence between objective and subjective measures of financial well-being has not been properly addressed (Brüggen et al., 2017, p. 234).

2.1 | Research questions

Our study explored the following three research questions. First, to what extent could a composite measure of household financial ratios be employed to delineate distinct states of financial well-being? This was the central question in our study. Existing research measuring household finances utilizing income, expenditures, savings, and so forth, are unable to independently explain a significant portion of the variance of financial well-being. We intend to explore whether a combination of such measures in the form of simple financial ratios allow us to define financial well-being in a consistent manner. More details provided in the next section will explain the conceptual model utilized.

Second, what level of congruence exists between the identified typologies of financial well-being, which we have named the equilibrium model of the household (EMH), and the CFPB's Financial Well-Being Scale? Once the EMH identified typologies using household financial ratios, we evaluated these objective measures in terms of their correspondence to the CFPB scale. This was made possible since participants provided household financial data and completed the CFPB scale. Finally, can a systematic mapping of the stages of financial well-being onto the CFPB's Financial Well-Being Scale be achieved to establish mutually exclusive and comprehensive intervals of CFPB scores?

We contend that enhancing the utility of the CFPB's Financial Well-Being Scale's score for industry

practitioners could be achieved by identifying distinct ranges of values on this scale. Such delineations could potentially reflect specific conditions or areas for improvement in household finances, thereby increasing the actionable insights derived from CFPB scores. Based on the research questions outlined in this section, the following hypotheses were created:

H1a. No combination of financial ratios can reliably identify distinct states of financial well-being.

H1b. There is at least one combination of financial ratios that can reliably identify distinct states of financial well-being.

H2a. The typology of financial well-being based on household financial ratios does not align with CFPB scores.

H2b. The typology of financial well-being based on household financial ratios is directly aligned with CFPB scores.

H3a. The typology of financial well-being based on the household financial ratios does not result in nonoverlapping, comprehensive mapping of the CFPB scale.

H3b. The typology of financial well-being based on the household financial ratios results in nonoverlapping, comprehensive mapping of the CFPB scale.

2.2 | Conceptual model

We defined typologies of household financial well-being using self-reported data from a small random sample of individuals in the United States. Using this dataset, we calculated four critical financial ratios. These ratios measure households' success in managing a budget, generating a positive cash flow, servicing household debt, and maintaining adequate levels of liquidity. By combining these ratios, we created an ordinal ranking of stages of financial well-being. Defining stages along the proverbial financial well-being continuum could be a valuable tool in the toolbox of any financial coach or counselor. These categories can typify common financial issues. Then counselors can offer options to help families move along the financial well-being continuum toward higher levels of financial well-being and upward mobility. Key aspects of our conceptual model, along with details regarding the

ratios and the process of creating the categories, are discussed in the next section.

2.2.1 | Equilibrium model of the household

Households that attain financial success often do so through the gradual accumulation of wealth. This accumulation process typically hinges on the judicious access to and utilization of credit. However, for the majority of households, the ability to access credit and build wealth is contingent upon the generation of income and the establishment of a surplus of income over expenses. This framework, termed by the authors as the EMH, suggests that the journey toward the apex of financial well-being commences with the establishment of a consistent income stream and the capacity to maintain a flexible monthly budget without being unduly burdened by excessive debt. This combination creates the environment for households to increase their savings rate, a factor already determined to be imperative to financial well-being.

Within the EMH framework, a household can flourish by generating a surplus of cash flow, which, when coupled with access to credit, serves to bolster the household's net asset position. Moreover, the success of households in accumulating wealth is contingent upon their adeptness in managing financial uncertainties by maintaining sufficient liquidity to mitigate potential risks. The EMH framework delineates four pivotal financial ratios and establishes a hierarchy among these ratios to define various stages of household financial well-being.

Household financial ratios

Financial ratios can measure certain aspects of the financial health of the household. A study by Griffith (1985) is one of the early writings on this topic. Subsequent efforts by other scholars, including a widely referenced article by Greninger et al. (1996), identified several important financial ratios and benchmarks for household financial well-being. DeVaney (1994), argued that household financial ratio analysis serves three purposes; to objectively measure family finance, to track progress in the financial situation through time, and to help financial educators, counselors, and planners guide families.

Harness et al. (2008) provided a comprehensive review of the literature on household financial ratios and indicators. The authors noted that, in most cases, the list of financial indicators identified by various researchers includes five or more items. However, indicators such as solvency ratio, liquidity ratio, debt-to-income ratio, investment assets ratio, and savings ratio appeared to be

the common denominators among the scholarly writings on this topic (ArthaYantra, 2013; Garman & Fargue, 2012; Kim & Lyons, 2008; Yao et al., 2003).

Measuring household financial ratios remain an important area of research in the field. Evidence has highlighted the inadequacy of household liquidity buffers amid the growing economic and income volatility that emerged in the aftermath of the Great Recession of 2008–2009 (Gjertson, 2015; Lusardi et al., 2011; Morgan Chase & Co. Institute, 2015). In response to this evidence, calls to create tools and policies that help households improve their ability to organize and manage bottom lines appeared in several research articles and/or reports (ibid).

Benchmarks

Utilization of household financial ratios provides a convenient method to evaluate household financial well-being. However, in building a new financial well-being scale using these ratios, complications may arise unless these ratios are complemented by proper benchmarks or reference values. With few exceptions, there is no a priori knowledge of how different ranges of values of these ratios may relate to household financial well-being. Early studies looked at central measures, such as averages and medians of sample data. Different datasets were used to collect the necessary information for such analyses. In one example, the financial ratios recommended by Griffith (1985) were calculated for 3824 households from the 1983 Survey of Consumer Finances using a multistage probability sampling (Prather, 1990). A correlation analysis examined the relationships between age, income, and financial ratios. Another approach used expert opinion to create such benchmarks (Greninger et al., 1996). Using the Delphi method, Greninger et al. (1996) collected information from a panel of 156 financial planners and educators to identify reference values for household financial ratios. Later studies used correlational methods, including logistic regression and other multivariate classification models, to evaluate relevant benchmarks to explain one or more aspects of household financial well-being (DeVaney, 1994). Common measures or definitions of well-being included financial fragility, financial insolvency, and household financial satisfaction. Having a net worth of <1 month's income was used to define financial insolvency as a binary variable (1 = insolvent, 0 = solvent) (DeVaney, 1994). Using benchmark values, DeVaney also created binary-coded independent variables and set up a logistic regression model to assess the significance of each independent variable in explaining household financial insolvency. This approach offered a solution for creating or adjusting benchmarks.

TABLE 1 Common financial indicators and recommended benchmarks.

Indicator	Definition
Estimated Monthly Savings Rate (EMS)	Defined as the difference between monthly net income and monthly expenses as a share of monthly gross income. Guideline: 1. 10% recommended by Greninger et al. (1996). 2. Benchmark value of 0 used in this study to separate households who are able to manage a monthly budget from those who do not.
Percentage of Fixed Expenses (PFE)	This indicator measures the proportion of fixed expenses to the net monthly expenses of the household. Guidelines: 1. No specific benchmarks exist for this indicator. This study assumes a value $\leq 65\%$ to be consistent with the idea of a healthy budget composition.
Credit Insolvency Ratio (CIR)	This is a variant of debt service to income ratio. It measures the proportion of monthly nonmortgage payments to net monthly income. Guidelines: 1. Less than 15% by Greninger et al. (1996).
Average Liquidity Ratio (ALR)	This ratio measures the amount of liquid assets (cash and near cash assets) relative to monthly expenses. Guidelines: 1. Between 200% and 600% by Griffith (1985) or 300% recommended by Greninger et al. (1996). 2. 250% if after-tax income is used in the denominator per DeVaney (1994) 3. A benchmark value of 200% is used in this study

Financial ratios defining financial well-being

Previous studies looked at the relationship between aspects of financial well-being and household financial ratios. For example, Garrett and James III (2013), looked at the relationship between household financial satisfaction and financial ratios. A study by Brunetti et al. (2012) looked at the role of household portfolio composition on financial fragility in Italy. One interesting aspect of this study is the classification of households into four nominal categories based on two general conditions: (1) whether income exceeded or was equal to planned household expenses, and (2) whether liquid household

assets were sufficient to meet potential unexpected expenses. The resulting categories were good financial position, financially fragile, dissaving, and financially constrained. Households were assumed to be in a good financial position if they met both conditions. The household was considered financially fragile if it met the first condition but not the second. If the household did not meet the first condition but met the second, the household was labeled as dissaving, over-consuming but liquid. Finally, the household was considered financially constrained if they did not meet either condition (Brunetti et al., 2012). The study used a multinomial logit model to associate household characteristics with the above-referenced classifications.

Another notable effort in this regard was the CFSI's Financial Health Segmentation Study which defined seven distinct segments within three overall tiers of financial health based on patterns of responses to a set of 24 objective and subjective questions (Gutman et al., 2015). Some financial ratios were included among these questions. The idea of creating categories of financial well-being was also suggested by Khashadourian (2016). He identified seven distinct categories of financial well-being. These ranged from financial distressed at the lowest level to upwardly mobile at the highest level. Khashadourian used a combination of values of household financial ratios within a data-assisted financial coaching network.

Selected financial ratios

Four financial ratios and indicators were selected for inclusion in our study. These variables are listed in Table 1. As used in the EMH, the Estimated Monthly Saving Rate (MSR) measures a household's ability to generate positive monthly cash flow. Percentage of Fixed Expenses (PFE) measures the flexibility of the household in managing a monthly budget which, when coupled with healthy levels of access to credit (CR), serves to bolster the household's net asset position. Moreover, the success of a household in accumulating wealth depends on its adeptness in managing financial uncertainties by maintaining adequate liquidity (ALR) in order to mitigate any potential financial risks, such as interruption of household income.

The EMH framework not only delineates four pivotal financial ratios but also established a hierarchy among these ratios to define various stages of household financial well-being (EMS $\rightarrow$ PFE $\rightarrow$ CIR $\rightarrow$ ALR). The hierarchy among financial ratios defined here posits that households strive to achieve financial equilibrium, first in the short run by balancing the household budget and generating a positive saving rate and then, in the long run, by managing household debt and liquid asset

accumulation. A sustainable positive cash flow will, throughout time, improve the ability of the household to manage debt and contribute to the accumulation of household assets resulting in long-term growth, eventually allowing the household to flourish financially.

2.3 | Significance of the study

The significance of this study is twofold. First, developing stages of financial well-being using financial ratios can make a strong case for content validity of the CFPB scale if the stages are shown to be consistent with the CFPB scale. Second, the study also examines whether stages of financial well-being may correspond to distinct intervals of the CFPB scale. Identifying such intervals would enhance the use of the CFBP measure by making it more actionable.

As an example of being more actionable, employers could use their employee CFPB scores, in combination with other types of employee data, such as salary records, and so forth, to identify necessary actions to improve employees' financial well-being (Bailey, 2019). In this context, stages of financial well-being may help identify areas of financial strength or weakness facing the household. Proper courses of action can then be taken to address these, thereby improving the effectiveness of financial capability programs offered by employers, universities, and community-based organizations. Finally, our approach can also help financial institutions in creating new products to improve the financial capability of their clients.

3 | METHODS

3.1 | Sampling and data collection process

To collect data, we developed an online survey consisting of two filtering variables, age and income. The survey also consisted of 40 multicomponent questions covering categories of a typical household budget and their assets and liabilities. Each participant also completed the CFPB survey. Capturing reliable and detailed household financial data through a survey proved to be complicated. We had to balance the amount of time needed to complete the survey against both the reliability of the collected data and the minimum amount of data elements required for a complete data analysis. Before collecting data for the main study, the research team spent nearly six months developing and refining the survey tool. A pilot study was conducted in January 2022 to evaluate the quality of the collected data. The research team also

recruited four volunteers who took the survey in the presence of the research team enabling communication with the volunteers during administration. Several recommendations led to the refinement of the survey tool, such as changing responses to certain data fields from "required" to "optional."

Power analysis was conducted using the G-Power software to estimate the required sample size to the main survey. A total sample size of 504 cases was estimated for a one-way, fixed effects analysis of variance (ANOVA) with six groups, using $\alpha = 0.05$, power $(1 - \beta) = 0.95$, and minimum effect size of $n^2 = 0.2$. The main survey was launched by our data partner, Qualtrics, in early April 2022, and closed mid-May 2022. A total of 506 responses from across the United States were collected. Initial inspection of the collected data resulted in the elimination of 90 records from the sample. Records were eliminated if sections of the survey were left completely blank or if the records included inconsistent or distorted data, such as entering a value of "1" for all data fields of the household budget. While losing $\sim 18\%$ of the original sample was not desirable, given the detailed nature of the survey, it was neither unexpected nor unprecedented. For example, in the CFSI study sample of 7152 cases, more than 31% of the sample was excluded when estimating the complete ordered Logit model where financial ratios were also included (Gutman et al., 2015, p. 34).

It should be emphasized that financial ratios may matter differently among the various stages of the life-cycle. For example, the ability to borrow and/or accumulate wealth may matter more at a younger age than later in life. Additionally, the ability to take reasonable action to improve financial well-being may be limited for both younger and older individuals. For this reason, the age filter constrained completion to individuals between the ages of 25 and 60. The income filter provided a proportional sample of responses for three categories of income; <\$50,000 per year (30% of the sample), between \$50,000 and <\$100,000 per year (50% of the sample), and over \$100,000 per year (20% of the sample). More details on the financial and demographic results are provided in Table 3.

The research team initially expected a participant to complete the survey in ~ 20 min while experiencing up to 15% missing/bad records. On average, participants spent over 18 min completing the survey. The survey was taken anonymously. At the end of the survey, respondents received a brief description of each item.

The remaining 416 records were subjected to a secondary review for data consistency and correction. Several records required minor corrections. Most of the important corrections were related to data frequencies. Our survey asked for income, taxes, and budget items to

TABLE 2 Ordinal Rankings of household financial status in the equilibrium model of the household framework.

Cashflow ^a	Budgeting	Borrowing	Liquidity	Rank
EMS > 0	PFE ≤ 65	CIR ≤ 15	ALR ≥ 200	1
EMS > 0	PFE ≤ 65	CIR ≤ 15	ALR < 200	2
EMS > 0	PFE ≤ 65	CIR > 15	ALR ≥ 200	3
EMS > 0	PFE ≤ 65	CIR > 15	ALR < 200	4
EMS > 0	PFE > 65	CIR ≤ 15	ALR ≥ 200	5
EMS > 0	PFE > 65	CIR ≤ 15	ALR < 200	6
EMS > 0	PFE > 65	CIR > 15	ALR ≥ 200	7
EMS > 0	PFE > 65	CIR > 15	ALR < 200	8
EMS ≤ 0	PFE ≤ 65	CIR ≤ 15	ALR ≥ 200	9
EMS ≤ 0	PFE ≤ 65	CIR ≤ 15	ALR < 200	10
EMS ≤ 0	PFE ≤ 65	CIR > 15	ALR ≥ 200	11
EMS ≤ 0	PFE ≤ 65	CIR > 15	ALR < 200	12
EMS ≤ 0	PFE > 65	CIR ≤ 15	ALR ≥ 200	13
EMS ≤ 0	PFE > 65	CIR ≤ 15	ALR < 200	14
EMS ≤ 0	PFE > 65	CIR > 15	ALR ≥ 200	15
EMS ≤ 0	PFE > 65	CIR > 15	ALR < 200	16

^aEMS categories are defined based on the condition of a free cashflow of ≥3% of total monthly expenses for a value of 1 and 0 otherwise.

be reported based on their corresponding monthly values. However, in some cases, values were reported annually. For example, some respondents reported monthly income that did not match the value of the income filtering variable they had previously selected.

3.2 | Data collection

Participants in our study were asked to complete the CFPB survey. In addition, participants were asked to provide household financial information, including income, taxes paid, typical expenses, asset values, and liability amounts. We used granular data from our sample to calculate the following four financial ratios:

$$EMS = \frac{MNI - TME}{MGI} \times 100, \quad (1)$$

$$PFE = \frac{FE}{MNI} \times 100, \quad (2)$$

$$CIR = \frac{MP_0}{MNI} \times 100, \quad (3)$$

$$ALR = \frac{LNLA}{TME} \times 100. \quad (4)$$

In these formulas, MNI is the net (after-tax) monthly income, TME measures the total monthly household

expenses, MGI is the monthly gross (before tax) income, FE is a measure of fixed expenses in the household budget, MP_0 measures all nonmortgage monthly payments on existing debt, and finally, LNLA measures all liquid and near-liquid household assets, including money in checking and savings accounts, college savings, retirement accounts, and other financial investments.

3.2.1 | Categorizing household financial well-being based on financial ratios

To define categories of household financial well-being, we used the indicators and their corresponding benchmark values listed in Table 1 to create four binary variables, each with categories 0 or 1 representing values below or above the stated benchmark. However, in reference to EMS, we made an exception by using a slightly different condition:

$$\text{Condition}_1 = \begin{cases} \text{if } (MNI - TME) \leq 0.03 * TME = 0 \\ \text{if } (MNI - TME) > 0.03 * TME = 1 \end{cases} \quad (5)$$

In other words, we replaced EMS with a measure of household cash flow such that a value of 1 is assigned if the monthly free cash flow of the household is positive and >3% of the monthly household expenses. Since we rely on self-reported data, using a 3% threshold for free cash flow offers the added confidence that any potential

under-reported monthly expenses had a smaller chance of misclassifying households based on positive versus negative cash flow positions.

Next, we combined these four binary variables according to the EMH hierarchy to identify 2⁴ or 16 possible ordinal ranks of financial status as given in Table 2. To better understand the ordinal nature of these rankings, consider the following example. The financial status of a household in Rank 4 (see Table 2) is decidedly higher than that of a family in Rank 10. The family in Rank 4 has a positive estimated savings rate (cash flow) and holds liquid assets that are sufficient to cover household expenses for at least 2 months in the event of a complete disruption of household income, unlike a family in Rank 10.

In general, our rankings follow households' behavior in terms of their ability to save (i.e., estimated monthly savings rate, EMS, being either $>$ or ≤ 0 according to

Condition (1) above); flexibility in managing a budget (fixed expense ratio, PFE, being either $\leq$ or $> 65\%$), managing debt (credit insolvency rate, CIR, being either $\leq$ or $> 15\%$), and maintaining proper liquid asset levels (average liquidity rate, ALR, being either $\geq$ or $< 200\%$).

3.2.2 | Consolidation of categories using discriminant analysis

Attempting to categorize financial status into 16 distinct stages may compromise both practicality and logical coherence of the measure. For instance, while households with varying levels of liquidity likely have real financial differences, using strict thresholds, like an ALR ratio below or above 200%, might not capture these nuances effectively. Practicality required that a more condensed set

TABLE 3 Financial and demographic characteristics of respondents.

Category	Frequency (relative frequency)	Category	Frequency (relative frequency)
Where do you live?	416 (100%)	Racial/Ethnic background	416 (100%)
Major metropolitan area (very high-cost area)	72 (17.3%)	Asian/Pacific Islander	22 (5.3%)
Major city (high-cost area)	121 (29.1%)	Black or African American	123 (29.6%)
Small city (moderate-cost area)	132 (31.7%)	Hispanic or Latino	119 (28.6%)
Small town or rural area	91 (21.9%)	Native American or American Indian	6 (1.4%)
Household Composition	416 (100%)	Non-Hispanic White	133 (32.0%)
Living by myself (a single adult)	68 (16.3%)	Other	13 (3.1%)
Living with minors (I am the only adult in my household)	35 (8.4%)	Level of financial autonomy/responsibilities within the household	416 (100%)
Living with my spouse or life partner (no minors or other adults in the household)	101 (24.3%)	Financially independent. Also support other individuals outside immediate household	89 (21.4%)
Living with my spouse or life partner (including minors in the household)	114 (27.4%)	Financially independent	161 (38.7%)
Living with my spouse or life partner and other adults in the household (with or without minors)	43 (10.3%)	I share financial responsibilities with other member(s) of my household	135 (32.5%)
Living with other adults such as an adult son or daughter (with or without minors)	55 (13.2%)	Fully or partially dependent on other members of my household	26 (6.3%)
Highest level of education (any member of the household)	416 (100%)	Fully or partially dependent on people who are not part of my immediate household	5 (1.2%)
No high school diploma	4 (1.0%)	Employment status	415 (100%)
High school graduate, diploma, or GED	142 (34.1%)	Self Employed	49 (11.8%)
Associate degree	92 (22.1%)	Employed for wages	279 (67.2%)
Bachelor's degree	124 (29.8%)	Looking for work	27 (6.5%)
Master's degree or higher	54 (13.0%)	Not working and not looking for work	60 (14.5%)

Note: The bold distinguishes the survey question and the total number of responses in each demographic question.

of hierarchical categories be developed to define the various stages of financial well-being more appropriately.

To this end, we aimed for fewer and more broadly defined categories of financial well-being. Existing research suggested financial well-being scales consist of four to seven categories (Brunetti et al., 2012; Gutman et al., 2015; Khashadourian, 2016). To identify the optimal number of categories, first, we consolidated the ranks in Table 2 in different ways and developed three candidate variables, featuring between four and up to six categories of financial well-being. We used linear discriminant analysis (LDA), a mathematical maximization procedure, to identify the candidate variable with the optimal number of categories of financial well-being. LDA identified uncorrelated linear combinations of the independent variable, that is, the four financial ratios listed in Table 1, in such a way that results in maximum separation between the categories, or stages, of financial well-being. LDA was conducted three times, each time using one of the above-referenced candidate variables as the dependent variable. Discriminant functions can be defined in the following way,

$$DF_{ij} = \alpha_{ij} + \beta_{1i}Z_{1j} + \beta_{2i}Z_{2j} + \beta_{3i}Z_{3j} + \beta_{4i}Z_{4j}, \quad (6)$$

where DF_{ij} is the i th discriminant score for household j , Z_{kj} , $k = 1, 2, 3, 4$ is the k th financial ratio for household j and β_{ki} , $k = 1, 2, 3, 4$ is the k th coefficient estimated for the i th discriminant function. It is important to note that in a discriminant model with k independent variables and p categories of the dependent variable, the maximum number of discriminant functions is always the smaller of $(p - 1)$ and k (Mertler et al., 2022). Group centroids were then defined for all categories of financial well-being for each of the i discriminant functions to form the mean vector for each category. These vectors were the reference values of each group in the discriminant space and were used for classification purposes (Tabachnick & Fidell, 2013).

Our primary goal in utilizing LDA was to evaluate the accuracy of the model in classifying cases into categories, or stages, of financial well-being. Since we created three candidate variables with a different number of categories of financial well-being, our criteria to select the best categorical variable consisted of (i) significance of all discriminant functions and (ii) the overall classification accuracy, or hit rate, of the model.

3.2.3 | Contrasting financial well-being categories to CFPB scores

Building on the identified financial well-being stages based on household financial ratios, we addressed the

second research question. This question explored the relationship between these stages and household scores on the CFPB Financial Well-Being Scale. Using the General Linear Model in SPSS version 29, a simple one-way ANOVA determined whether significant differences exist in CFPB scores across categories of financial well-being. Additionally, post hoc comparisons helped evaluate the existence of significant differences in two-by-two comparisons, leading to the establishment of confidence intervals for CFPB scores within each category. Given the ordinal nature of the stages of financial well-being, we expected upper categories of financial well-being to correspond to higher values of the CFPB scores.

4 | RESULTS

Table 3 highlights some of the demographic characteristics of the respondents. Our sample consisted of 416 participants from across the US Geographically, the sample was diverse with the highest concentration of responses coming from Texas (12.7%), California (10.8%), New York (9.6%), and Florida (8.4%). The majority of respondents identified as female (64.4%). The remaining respondents identified as male (35.1%) or as another gender or preferred not to answer (0.04%). The average household size was 2.98 persons with a standard deviation of 1.51. On average, there were 2.13 adults per household (standard deviation = 0.0907) and 1.58 jobs, full-time or part-time, among household members (standard deviation = 0.0958). Refer to Table 3 for a more comprehensive breakdown of the sample's demographics, including factors such as location type, race/ethnicity, household composition, financial autonomy of the respondent, education level, and employment status.

4.1 | Research question one: Extracting a financial well-being scale using discriminant analysis

In our study, direct discriminant analysis was applied to the financial ratios listed in Table 1 to determine the optimal number of categories for a proposed financial well-being scale. To address outliers, variables CIR and ALR were logarithmically transformed. Candidate variables featured four, five, and six categories based on consolidated ranks from Table 2. A seven-category variable was not considered due to uneven frequency distribution across categories.

We evaluated three discriminant models, each using a candidate variable of financial well-being as the categorical dependent variable, with the above-mentioned financial ratios as the independent variables across all models.

Discriminant function	Eigenvalue	% of variance	Canonical correlation
Model 1 (six categories)			
1	2.246	55.4	0.832
2	1.582	39.0	0.783
3	0.217	5.3	0.422
4	0.010	0.2	0.098
Model 2 (five categories)			
1	1.991	57.4	0.816
2	1.241	35.8	0.744
3	0.236	6.8	0.437
4	0.003	0.1	0.052
Model 3 (four categories)			
1	1.707	61.0	0.794
2	0.884	31.6	0.685
3	0.210	7.5	0.416

TABLE 4 Eigenvalues and canonical correlations.

	Wilk's Lambda	Chi-square	df.	Sig.	Overall hit rate
Model 1 (six categories)					74.4%
1 through 4	0.097	932.752	20	<0.001	
2 through 4	0.315	461.816	12	<0.001	
3 through 4	0.814	82.317	6	<0.001	
4	0.990	3.849	2	0.146	
Model 2 (five categories)					73.5%
1 through 4	0.120	849.974	16	<0.001	
2 through 4	0.360	410.072	9	<0.001	
3 through 4	0.807	86.069	4	<0.001	
4	0.997	1.075	1	0.300	
Model 3 (four categories)					81.3%
1 through 3	0.162	731.562	12	<0.001	
2 through 3	0.439	331.176	6	<0.001	
3	0.827	76.530	2	<0.001	

TABLE 5 Test of discriminant function and classification overall hit rate.

Tests of equality of group means, not reported here, indicated significant variation in mean values of independent variables across financial well-being categories. For models with five or six category dependent variables, Model 1 and Model 2, four discriminant functions were extracted, while three functions were extracted for the four-category model, Model 3. Table 4 presents eigenvalues and canonical correlations of discriminant functions. In all models, the first discriminant function exhibited a strong relationship with financial well-being categories. In addition, test of discriminant functions (see Table 5) revealed that all functions were significant in Model 3, but not in Model 1 nor Model 2. Significance of all functions is crucial for model selection, indicating

appropriate discrimination based on all financial ratios (Mertler et al., 2022).

Based on the discriminant results, we selected Model 3, which identifies four distinct stages of financial well-being (see Table 7), as the optimal choice of categories of financial well-being. In this model, all discriminant functions, including the last one, were significant (Wilk's = 0.827, $X^2(2) = 76.52$, $p < 0.001$), implying strong discriminatory power. Model 3 also explained the highest variance, 61%, based on the first discriminant function and achieved a classification hit rate of 81.3%, which was higher than the other two models. Table 6 presents the structure matrix of correlations between the four financial ratios and the discriminant functions

for Model 3. Looking at the first discriminant function, the structure matrix revealed that correlations are aligned with the expected signs and strength per the EMH.

4.2 | Research question two: Contrasting CFPB score to financial well-being categories

To answer how our objective definition of financial well-being compared with CFPB's Financial Well-Being Scale, we evaluated the existence of differences in CFPB's scores across four stages of financial well-being as defined in Table 7. To conduct the ANOVA tests, we used the univariate General Linear Model in SPSS, version 29. Before conducting the test, data prescreening confirmed the assumption of homogeneity of variances across groups. The mean-based Levene statistic, $L(3,406) = 0.315$, $p = 0.0815$, resulted in retaining the null hypothesis of homogeneous variances. A similar result was obtained for the median-based statistic.

Test of normality of the distribution of CFPB scores within each group was evaluated using the Kolmogorov–Smirnov test which resulted in confirming the assumption of normality in Financially Flourishing, the first category, and Distressed, the last category. While the null hypothesis of normality was rejected for the two other groups, inspection of the skewness and kurtosis values for these groups revealed the overall symmetry of the

distribution in both cases. Both distributions, however, appeared slightly leptokurtic with kurtosis being slightly higher than one only in the Financially Stable, second category. No further action was taken to prescreen or transform the data (Mertler et al., 2022).

Results from the ANOVA indicated a significant main effect for the stages of financial well-being developed in our study, $F(3,406) = 32.99$, $p < 0.001$. Calculated effect size and the observed power of the test were also consistent with the ex-ante values set during the sample-size determination, $\eta^2 = 0.197$ and $(1 - \beta)_{\text{Obs.}} = 1.0$. Finally, Table 7 shows the mean and the standard deviation of the CFPB scores in each of the four categories. The number of observations in each category also appears in Table 7. As expected, mean CFPB scores consistently decline along the proverbial continuum of financial well-being according to the ordinal stages of financial well-being (see Figure 1). The mean CFPB score for the Financially Flourishing category was 63.08, falling to 51.31 for the Financially Stable category, and 48.76 for the Financially Fragile category. Finally, the Financially Distressed category had a CFPB score of 47.57. In reference to research question two, we concluded that the typology of financial well-being using financial ratios were consistent with the CFPB scale.

TABLE 6 Structure matrix of correlations between each variable and discriminant function (Model 3).

	DF1	DF2	DF3
ALR ^a	0.394	0.896 ^b	0.206
CIR ^a	−0.477	0.399	0.782 ^b
EMS	0.639	−0.312	0.696 ^b
PFE	−0.519	0.223	−0.533 ^b

^aLogarithmically transformed variable (base 10).

^bThe largest absolute correlation between the identified variable and any discriminant function (DF).

TABLE 7 Stages of financial well-being.

Stages of financial well-being	Rank consolidation	Frequency	Mean CFPB score	Standard deviation
Flourishing	1	93	63.08	11.329
Stable	2–5	140	51.31	11.537
Fragile	6–14	133	48.76	11.552
Distressed	15 and 16	44	47.57	13.005
All categories	—	410	52.75	12.961

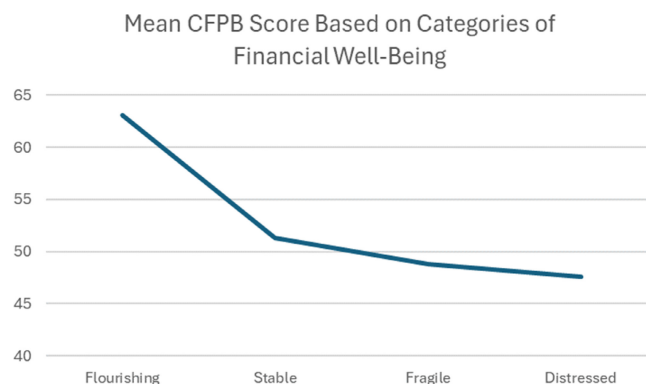


FIGURE 1 Estimated Marginal Means of CFPB Financial Well-Being Score.

TABLE 8 Confidence intervals around the mean, with and without inclusion of random effects.

Category	Mean	No random effects			With random effects		
		Standard error	Lower bound (95% CI)	Upper bound (95% CI)	Standard error	Lower bound (95% CI)	Upper bound (95% CI)
Flourishing	63.075	1.209	60.698	65.452	0.259	61.960	64.191
Stable	51.307	0.985	49.370	53.244	0.212	50.432	52.257
Fragile	48.759	1.011	46.772	50.747	0.217	47.827	49.692
Distressed	47.568	1.758	44.113	51.024	0.377	45.947	49.190

4.3 | Research question three: Mapping categories of financial well-being onto CFPB scale

While the results from the ANOVA were statistically significant and category averages declined steadily from the highest category of Financially Flourishing to Financially Distressed, relatively high standard errors within these categories made it impossible to define nonoverlapping confidence intervals for all four distinct states of financial well-being (see Table 8). Multiple comparison post-hoc tests were conducted to determine which category of financial well-being had average CFPB scores that were significantly different from other categories. Results, not reported here, indicated that differences between all categories were statistically significant, at least at $\alpha = 0.1$, except for the pairwise comparison between Financially Fragile and Financially Distressed states.

Estimated marginal means and their corresponding 95% confidence intervals in Table 8 presented how the intervals for the three lower categories of financial well-being overlap. While partially overlapping categories, such as the overlap between Financially Stable and Financially Fragile, was not a significant concern in terms of practicality and identification of states of financial well-being, the same cannot be claimed for the Financially Distressed and Financially Fragile stages. In this case, the upper bound of the Financially Distressed category was greater than the upper bound of the Financially Fragile category, thus completely masking the Financially Fragile category.

Part of the issue of overlapping categories could be attributed to large random effects in the sample data (Seltman, 2012). We ran the ANOVA model again by including the random effects of the survey participants. Inclusion of the random effects lowered the within-group variability and improved the group-level effects, resulting in significant improvements in pairwise comparisons. All pairwise comparisons in the model that included the random effects were statistically significant at the 5% level except for the last two groups. In addition, the updated

model provided nonoverlapping categories for the first three groups (see Table 8). In reference to research question three, our study was not successful in identifying an exhaustive and nonoverlapping ranges of financial well-being categories along the CFPB scale.

5 | DISCUSSION

Our results suggest that financial ratios, considered an objective measure, can effectively categorize financial well-being. In addition, these categories are consistent with the CFPB scale, which we consider to be a subjective measure. This implies the CFPB scale might capture objective financial behavior to a greater extent than previously thought. We contend that our results enhance the validity of the CFPB survey as a simple and practical way to measure financial well-being. We arrived at this conclusion because the EMH objective measure of financial well-being introduced in our study was developed independently from the CFPB scores. Statistical tests not only implied that significant differences existed across categories of financial well-being, but that declining CFPB average scores in each category were found to be in line with the expected ordering of such categories along the EMH financial well-being continuum.

A second objective of the study, defining nonoverlapping and exhaustive categories of financial well-being, was not supported since results were inconclusive across the entire domain of possible CFPB scores. On the one hand, confidence intervals for the two categories of Financially Distressed and Financially Fragile had significant overlaps. On the other hand, the intervals for the two middle categories, Financially Fragile and Financially Distressed, while not overlapping, appeared to be too narrowly defined given the high variability of the CFPB scores. Several factors may contribute to this problem, but four obvious ones include (i) the multifaceted nature of financial ratios in influencing financial well-being, (ii) measurement and calculation errors, (iii) small sample size, and (iv) high standard deviation of the

distribution of the CFPB scores. We discuss these items in the next two sections.

5.1 | Implications

One argument that challenges the idea of utilizing financial ratios, as part of the EMH, to define financial well-being relates to the multidimensional influence of financial ratios on financial well-being of individuals in different stages of life. As an example, higher levels of indebtedness of a young individual because of student loans may not necessarily have the same effect on financial well-being when compared with a high level of indebtedness of another person who carries large amounts of credit card debt. It is even plausible to assume that the financial well-being of an individual who has student loans to pay off is higher *because* student loans imply investing in higher education, which could offer better employment opportunities later in life. However, this argument, while true, does not necessarily undermine the use of financial ratios as an objective measure to define financial well-being. This is because a higher level of indebtedness unequivocally implies a lower state of financial well-being, everything else constant. In other words, the marginal effect of indebtedness on financial well-being is negative. Consider a student with only \$10,000 in student loan debt. This individual is better off compared with another student with \$100,000 in student loan debt, everything else equal. Still, the role of a financial ratio in determining financial well-being may be complex when analyzed in the context of other financial ratios. In our study, this is clearly demonstrated in the structure matrix of the discriminant functions. In the first discriminant function, indebtedness appears with the expected negative sign (see Table 6). Yet the same variable has the highest correlation with the third discriminant function in which it has a positive sign. When comparing other correlations in the third discriminant function, the positive sign may simply indicate a higher level of access to capital in connection with other financial ratios, leading to a higher level of financial well-being. We contend that this remains a true objective measure when the interpretation is made for the marginal effect.

5.2 | Limitations

Measurement errors in financial ratios can result in multiple complications related to the definition of a categorical financial well-being scale, especially when self-reported data is used for this purpose. Nevertheless,

nonsystematic measurement errors and missing values are inevitable in any granular sampling of household micro statistics. In addition, we relied on self-reported data simply because direct observation of data is almost impossible to implement. In a handful of examples in the literature where data is collected through observation of bank account activities and transactions, we suspect that any reduction in nonsystematic errors may have been achieved at the expense of elevating the probability of systematic errors in data collection and calculation. As an example, when data is collected from a single financial institution, the implicit assumption is that all financial transactions of a household are conducted in a noncash form and through a single financial intermediary.

Our sample size was also relatively small compared with similar studies. This was primarily due to the consideration of the effect size in estimating the required sample size. Effect size remains an essential element in evaluating the results of an ANOVA model. Technically speaking, one may select an arbitrarily large sample size to find significant group differences at the expense of lowering the effect size, everything else held constant (Mertler et al., 2022; Nolan & Heinzen, 2016). However, since our study intends to define categories of financial well-being that are meaningfully distinct from one another, adherence to a minimum effect size of close to 0.2 seems reasonable, even though it resulted in calculating a significantly smaller sample size for our study. We want to clarify that although our initial calculation of the necessary sample size was based on a six-category factor, the discriminant analysis ultimately identified a factor with only four distinct categories. This reduction in the number of categories helped compensate for any decrease in observed power resulting from the exclusion of 18% of the original data due to missing or inconsistent data.

While our inability to define nonoverlapping and exhaustive stages of financial well-being in research question three could partly be the result of measurement errors in self-reported data or reliance on a small sample size, we suspect that the high standard deviation present in the distribution of the CFPB scores could also contribute to this problem, as high standard deviation may indicate the existence of noise in the data. Evaluation of the mean and standard deviation of the CFPB score in the CFPB's national data sample, consisting of 6389 observations, yields a coefficient of variation of around 25% (CFPB, 2017). We obtained a similar result, 24%, in our sample. An evaluation of this claim could be a topic for future research. For example, one way to reduce the degree of variability of data in such cases is to properly partition the data based on subgroups that may represent

TABLE 9 Description of financial well-being categories within the EMH model based on household financial ratios.

Category	Description ^a	Corresponding CFPB range
Financially distressed	Household is facing negative cashflow and is struggling to manage the household budget due to high percentage of fixed expenses and excessive borrowing. The household may or may not have enough savings to last more than 2 months. Household will need to make long-term changes to monthly expenses to lower percentage of fixed expenses	Lowest to 46
Financially fragile	Fragile households may or may not be impacted by excessive borrowing, or insufficient savings. However, these households typically belong to one of the following two categories 1. Unable to generate a positive cashflow but maintains a flexible budget allowing the household to improve cash flow position 2. Generates a positive cashflow, but faces challenges in maintaining a flexible budget, increasing the likelihood of moving into a negative cash flow position Short-term changes to household budget are required to improve estimated savings and maintain a flexible budget	47–49
Financially stable	Stable households generate a positive cashflow. In most cases the household is able to manage a flexible budget. Some households may not be able to manage a flexible budget, but in such cases, they are still able to keep household borrowing at a manageable level while maintaining enough savings to last for more than 2 months. Household should pay attention to maintain CIR at or below 15%	50–61
Upwardly mobile	Upwardly mobile households generate a positive cashflow. They have flexible budgets and maintain borrowing at manageable levels while holding savings enough to last for more than 2 months. Liquid savings enough to cover 6 months' worth of expenses is desirable for this group	62 to highest

^aDescriptions are derived based on the consolidated categories of financial well-being as reported in Table 7.

distinct populations of respondents. The existing CFPB scale already incorporates a two-way partitioning of data; one, based on age (separating responses obtained from people 62 years of age and older), and two, according to how the survey is administered (self-administered or administered by others) (CFPB, 2015). In the 2017 national study, the CFPB defines six broad subgroups of participant characteristics and studies the components of each subgroup in terms of their relationship to the calculated CFPB score. The report concludes that “...there seems to be no single set of characteristics or circumstances that lead an individual to have higher or lower financial well-being” (CFPB, 2017, p. 20). In other words, the research team at the CFPB did not find strong reasons to partition the dataset further to improve, or lower, the variance of the CFPB scores. We find this description imprecise because it draws its conclusion solely based on direct comparisons and without considering the possibility of multidimensional interaction between characteristics. Accordingly, we cannot rule out the possibility of the presence of a disproportionately high level of noise in the CFPB data due to incomplete partitioning. The noise in the data does not undermine the overall validity of the CFPB Financial Well-Being Scale. However, it may cast some doubt on the reliability of the measure and necessitates further research on the topic.

5.3 | Future research

Another area that might be of interest to researchers is the relationship between the emotional and mental state of an individual and their perception of financial well-being (Agarwal & Mazumder, 2013; Comerton-Forde et al., 2022; Das, 2019). In a recent study, Enete et al. (2022) argued that negative and positive emotions influence individual perceptions of financial well-being in negative and positive ways, respectively. Another study suggests that financial well-being may mean different things to people in different age groups (Riitsalu et al., 2024). The CFPB did not include the mental and emotional state of the respondents among the characteristics studied in the six subgroups, nor did it partition the data for age subgroups other than above or below 62 years of age. Results from these and similar studies may offer additional reasons to explore the idea of improving data partitioning for better results. In the meantime, we propose using the exhaustive range of values presented in Table 9 as a starting point for future studies in this regard. In fact, by using the range of value in Table 9 along with their corresponding descriptions, financial coaches and other personal finance professionals can provide more practical recommendations to their clients on how to improve their financial well-being.

6 | CONCLUSION

We conclude this article by recommending that further research look at ways to reduce the noise in the CFPB data. We also believe that any future attempt to explore the research questions in our study should be carried out using separate datasets. Defining stages of financial well-being, as in the EMH, would benefit from having a larger data sample to improve the classification and categorization of financial well-being. At the same time, contrasting CFPB scores to the EMH categories of financial well-being and attempting to define nonoverlapping ranges of CFPB scores requires strict adherence to a minimum acceptable effect size, which limits the desired sample size.

In addition, a significant contribution of this research is not only to support prior studies results related to the importance of household savings rates on financial well-being, but also the progressive interaction of the four financial ratios on the movement along the financial well-being continuum. The initial building block to financial well-being is a positive savings rate followed by no more than 65% of a household budget composed of fixed expenses. Add to these first two, a household with <15% of their income dedicated to repaying nonmortgage debt, moves that household to a higher stage of financial well-being. The last factor, maintaining at least 200% of monthly expenses in liquid assets, moves the household to the Financially Flourishing stage. Also of note is the congruence between perceived financial well-being and the behaviors that either precede or follow that perception. Knowledge of this progressive interaction, combined with the CFPB score, can enhance the effectiveness of financial counseling efforts in the United States.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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